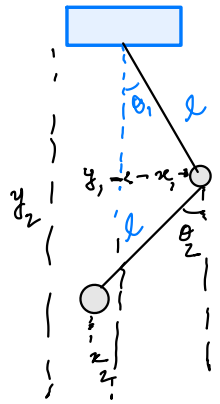
Tutorial on 12/ Nov / 2025

Let's look at double pendulum



$$T = \frac{1}{2} 2m\, l^2 \dot{\theta}_1^2 + \frac{1}{2} m l^2 \dot{\theta}_2^2 + m l^2 \cos(\theta_1 - \theta_2)\, \dot{\theta}_1 \dot{\theta}_2$$

$$V = -2mgl\cos\theta_1 - mgl\cos\theta_2$$

Compute the modes for small oscillation about equilibrium point.

(*) To find the equilibrium points set $\frac{\partial V}{\partial \theta_1}$ & $\frac{\partial V}{\partial \theta_2} = 0$

$$\frac{\partial V}{\partial \theta_1} = 2mgl\sin\theta \quad \& \quad \frac{\partial V}{\partial \theta_2} = mgl\sin\theta_2$$

Stable equilibrium points are at $\theta_1 = \theta_2 = 0$

Let $\theta_1 = 0 + \eta_1$ & $\theta_2 = 0 + \eta_2$

$$T = ml^2\dot{\eta}_1^2 + \frac{1}{2} m l^2 \dot{\eta}_2^2 + m l^2 \cos(\eta_1 - \eta_2)\, \dot{\eta}_1 \dot{\eta}_2$$

$$V = -2mgl\cos\eta_1 - mgl\cos\eta_2$$

$\eta_1 \ll 1$ and $\eta_2 \ll 1$

$$\cos\eta_1 \simeq 1 - \frac{\eta_1^2}{2} \quad \& \quad \cos\eta_2 \simeq 1 - \frac{\eta_2^2}{2}$$

$$\cos(\eta_1 - \eta_2) \simeq 1 \qquad \eta_1 - \eta_2 \simeq 0$$

with this we get

$$T \simeq ml^2\dot{\eta}_1^2 + \frac{1}{2} m l^2 \dot{\eta}_2^2 + m l^2 \dot{\eta}_1 \dot{\eta}_2$$

$$V \simeq -3mgl + mgl\,\eta_1^2 + \frac{1}{2} mgl\,\eta_2^2$$

$$T = \begin{pmatrix} 2ml^2 & ml^2 \\ ml^2 & ml^2 \end{pmatrix} \quad \& \quad V = \begin{pmatrix} 2mgl & 0 \\ 0 & mgl \end{pmatrix}$$

The EOM is $\det|V - \omega^2 T| = 0$ $\quad \begin{vmatrix} 2ml(g - l\omega^2) & -ml^2\omega^2 \\ -ml^2\omega^2 & ml(g - l\omega^2) \end{vmatrix} = 0$

$$\Rightarrow \quad 2(g - l\omega^2)^2 - l^2\omega^2 = 0$$

$$\omega^4 l^2 - 4\omega^2 g l + 2g^2 = 0$$

$$\omega_1^2 = \frac{4gl + \sqrt{16g^2l^2 - 8g^2l^2}}{2l^2}$$

$$\omega_1^2 = \frac{4gl - \sqrt{16g^2l^2 - 8g^2l^2}}{2l^2}$$

$$\omega_1^2 = \frac{g}{l}(2+\sqrt{2}) \quad \& \quad \omega_2^2 = \frac{g}{l}(2-\sqrt{2})$$

are the modes of oscillation

⊛ one can also determine the amplitude.
That is for home!

Problem 2 Find the oscillation frequency & Amplitude for small motion about equilibrium point

$$T = \frac{1}{2}m_1\dot{x}_1^2 + \frac{1}{2}m_2\dot{x}_2^2 \quad \text{where } m_1 = 2m \ \& \ m_2 = m$$

and

$$V = \frac{1}{2}k_1x_1^2 + \frac{1}{2}k_2(x_2 - x_1)^2 + \frac{k_3}{2}x_2^2 \quad \text{and} \quad k_1 = 4k \quad k_2 = k \quad k_3 = 2k$$

The equilibrium point is at $x_1 = x_2 = 0$

and for small motion η_1 & η_2 about equilibrium we have

$$T = \frac{1}{2}m_1\dot{\eta}_1^2 + \frac{1}{2}m_2\dot{\eta}_2^2 \Rightarrow T = \begin{pmatrix} m_1 & 0 \\ 0 & m_2 \end{pmatrix}$$

$$V = \frac{1}{2}k_1\eta_1^2 + \frac{1}{2}k_2(\eta_2 - \eta_1)^2 + \frac{k_3}{2}\eta_2^2$$

$$V = \begin{pmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_3 + k_2 \end{pmatrix} = \begin{pmatrix} 5k & -k \\ -k & 3k \end{pmatrix}$$

The EOM is given by

$$\det\left| V - \omega^2 T \right| = 0$$

$$\begin{vmatrix} 5k - 2m\omega^2 & -k \\ -k & 3k - m\omega^2 \end{vmatrix} = 0$$

$$(5k - 2m\omega^2)(3k - m\omega^2) - k^2 = 0$$

$$2m^2\omega^4 - 11\,km\omega^2 + 14\,k^2 = 0$$

we get soln for $\omega_1^2 = \frac{2k}{m}$ & $\omega_2^2 = \frac{7k}{2m}$

find the Eigen vectors on your own!